T- Test

Aim:

To perform a one-sample *t-test* in Python to determine whether the average IQ score of a sample of students differs significantly from the population mean IQ of 100.

Procedure:

1. Import the necessary libraries — NumPy for data handling and SciPy for statistical testing.
2. Define the population mean (100) and the level of significance ($\alpha = 0.05$).
3. Collect or define the IQ scores of 25 randomly selected students as the sample data.
4. Use scipy.stats.ttest_1samp() to perform a one-sample *t-test* comparing the sample mean with the population mean.
5. Calculate the *t-statistic* and *p-value* using the test function.
6. Compute the sample mean using np.mean() to compare it with the population mean.
7. Print the sample mean, *t-statistic*, and *p-value* for interpretation.
8. Compare the *p-value* with the significance level ($\alpha = 0.05$).
9. If the *p-value* is less than α, reject the null hypothesis; otherwise, fail to reject it.
10. Conclude whether the sample mean IQ significantly differs from the population mean.

Program:

```
import numpy as np
from scipy import stats
population_mean = 100
alpha = 0.05
sample = np.array([98, 102, 101, 99, 100, 103, 97, 105, 99, 101,
                   104, 98, 100, 99, 102, 103, 97, 106, 100, 101,
                   98, 102, 99, 100, 104])
t_statistic, p_value = stats.ttest_1samp(sample, population_mean)
print("Sample Mean:", np.mean(sample))
print("T-Statistic:", t_statistic)
print("P-Value:", p_value)
if p_value < alpha:
    print("Reject the null hypothesis")
else:
    print("Fail to reject the null hypothesis")
```

Output:

```
Sample Mean: 100.72
T-Statistic: 1.4544217970979978
P-Value: 0.1587837609314068
Fail to reject the null hypothesis
```

Result:

Hence, a Python program was written to perform a one-sample *t-test*, and the result shows that there is **no significant difference** between the sample's average IQ score and the population mean IQ of 100 at a 5% significance level.